Fiscal Year:	FY 2025	**Task Last Updated:**	FY 01/30/2025
PI Name:	O'Banion, Kerry M.D., Ph.D.		
Project Title:	Effect of Space Radiation on Innate Immune System Homeostasis Drives Immune and Endothelial Cell Dysfunction and Neurodegeneration		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes (3) **Immune**:Risk of In Mission Impacts, Adverse Health Events or Long-Term Health Impacts due to Altered Immune Response		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	kerry_obanion@urmc.rochester.edu	**Fax:**	FY 585-756-5334
PI Organization Type:	UNIVERSITY	**Phone:**	585-275-5185
Organization Name:	University of Rochester		
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PI Web Page:			
City:	Rochester	**State:**	NY
Zip Code:	14642-0001	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019-2020 HERO 80JSC019N0001-HHCBPSR, OMNIBUS2: Human Health Countermeasures, Behavioral Performance, and Space Radiation-Appendix C; Omnibus2-Appendix D
Start Date:	02/01/2021	**End Date:**	01/31/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 01/31/2026 per the PI (Ed., 3/11/25).		
Key Personnel Changes/Previous PI:	Drs. Olschowka and Williams were removed from the grant based on their retirement. Dr. Tanzy's effort was reduced to 0 in April in line with our extended budget proposal.		

COI Name (Institution):	Calvi, Laura M.D. (University of Rochester) Love, Tanzy Ph.D. (University of Rochester) Olschowka, John Ph.D. (University of Rochester) Williams, Jacqueline Ph.D. (University of Rochester)
Grant/Contract No.:	80NSSC21K0542
Performance Goal No.:	
Performance Goal Text:	
Task Description:	The key objective of this proposal is to evaluate dose- and time-dependent effects of simulated space radiation on multiple tissue endpoints associated with normal aging. These endpoints include measures of the immune system, in particular those cells in the blood and tissues that represent a first line of defense for reacting to injury, the vascular compartment, another critical component of tissue homeostasis, and the brain, a critical system that shows age-associated degeneration which also occurs after radiation exposure. These studies will be carried out with male and female mice exposed to simulated space radiation at the NASA Space Radiation Laboratory using modern molecular techniques to measure tissue and cellular changes in gene and protein expression, alterations in cell phenotypes by flow cytometry, and behavioral tests and computer-assisted imaging in brain to quantify degenerative changes. This proposed work responds to Topic 6, Space Radiation and Tissue Homeostasis, in NASA's solicitation, and will provide key information about space radiation effects on the promotion of age-associated changes in multiple systems that are likely tied together in the etiology of late degenerative changes that may impact future astronaut health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Research focuses on mechanisms of normal tissue injury in brain and bone marrow following radiation exposure. Identification of specific pathways involved in late effect of radiation on these tissues may provide clues for protective or mitigating strategies in the setting of clinical radiotherapy or accidental radiation exposure.
Task Progress:	In this fourth year of the grant, we carried out irradiations during NASA Space Radiation Laboratory-23C (NSRL-23C) for our fifth proposed experiments, which examines the effects of space radiation on mice carrying heterozygous deletions of the Idh2 gene in hematopoietic stem cells. For this experiment exploring clonal hematopoiesis, we irradiated 12 male and 19 female WT and Vav-Cre/IDHfl/+ mice with 100 cGy simGCRsim at BNL. An additional 9 male and 13 mice were similarly processed at the NSRL (e.g., placed in holders for similar times), but not exposed to radiation. Following irradiation all mice were shipped back to Rochester. All mice tolerated radiation and shipping. These mice are currently "aging" and will be examined for evidence of anemia and clonal hematopoieis by blood analysis over a time course as well as bone marrow analysis 1 year post-irradiation. Additional lines of mutant mice will be irradiated in 2025 to complete our planned experiments examining clonal hematopoiesis. We continue to carry out tissue and other analyses of measures collected from earlier cohorts of irradiated wild-type C57BL/6 mice. We presented abstracts describing brain and bone marrow finding projects at NASA Human Research Program (HRP) Investigators' Workshop (IWS) in February 2024 and January 2025, as well as scRNAseq results from irradiated microglia at the 2024 Annual Radiation Research Society conference. Manuscripts describing findings in the bone marrow compartment and the scRNAseq results are in preparation.
Bibliography Type:	Description: (Last Updated: 01/07/2026)
Abstracts for Journals and Proceedings	O'Banion MK, Calvi LM, Olschowka JA, Love TMT, Williams JP. "Effects of space radiation on innate immune system homeostasis drives immune and endothelial cell dysfunction and neurodegeneration." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Osabutey MA, O'Banion MK, Olschowka J. "Population-level analysis of microglial inflammatory states after low-dose space radiation exposure." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Osabutey MA, Olschowka J, O'Banion MK. "Investigating sex-dependent microglial transcriptional responses to galactic cosmic radiation." 70th Annual Radiation Research Society Meeting, Tucson, Arizona, September 15-18, 2024. Abstracts. 70th Annual Radiation Research Society Meeting, Tucson, Arizona, September 15-18, 2024. , Sep-2024
Abstracts for Journals and Proceedings	O'Banion MK, Calvi LM, Olschowka JA, Love TMT, Williams JP. "Effects of space radiation on innate immune system homeostasis drives immune and endothelial cell dysfunction and neurodegeneration." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Articles in Peer-reviewed Journals	Strohm AO, Johnston C, Hernady E, Marples B, O'Banion MK, Majewska AK. "Cranial irradiation disrupts homeostatic microglial dynamic behavior." J Neuroinflammation. 2024 Apr 3;21:82. https://doi.org/10.1186/s12974-024-03073-z ; PubMed PMID: 38570852; PubMed Central PMCID: PMC10993621 , Apr-2024
Articles in Peer-reviewed Journals	Strohm AO, Oldfield S, Hernady E, Johnston CJ, Marples B, O'Banion MK, Majewska AK. "Biological sex, microglial signaling pathways, and radiation exposure shape cortical proteomic profiles and behavior in mice." Brain Behav Immun - Health. 2025 Feb;43:100911. https://doi.org/10.1016/j.bbih.2024.100911 ; PubMed PMID: 39677060; PubMed Central PMCID: PMC11634995 , Feb-2025

Articles in Peer-reviewed Journals	Heneka MT, van der Flier WM, Jessen F, Hoozemanns J, Thal DR, Boche D, Brosseron F, Teunissen C, Zetterberg H, Jacobs AH, Edison P, Ramirez A, Cruchaga C, Lambert J-C, Laza AR, Sanchez-Mut JV, Fischer A, Castro-Gomez S, Stein TD, Kleineidam L, Wagner M, Neher JJ, Cunningham C, Singhrao SK, Prinz M, Glass CK, Schlachetzki JCM, Butovsky O, Kleemann K, De Jaeger PL, Scheiblich H, Brown GC, Landreth G, Moutinho M, Grutzendler J, Gomez-Nicola D, McManus RM, Andreasson K, Ising C, Karabag D, Baker DJ, Liddelow SA, Verkhratsky A, Tansey M, Monsonego A, Aigner L, Dorothée G, Nave K-A, Simons M, Constantin G, Rosenzweig N, Pascual A, Petzold GC, Kipnis J, Venegas C, Colonna M, Walter J, Tenner AJ, O'Banion MK, Steinert JR, Feinstein DL, Sastre M, Bhaskar K, Hong S, Schafer DP, Golde T, Ransohoff RM, Morgan D, Breitner J, Mancuso R, Riechers S-P. "Neuroinflammation in Alzheimer disease." Nat Rev Immunol. 2024 Dec 9. https://doi.org/10.1038/s41577-024-01104-7 , Dec-2024